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Effect of Indoor Temperature and Glazing with Saturated Color on Visual Perception of Daylight

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ABSTRACT

This paper presents the combined effect of indoor temperature (19 °C, 22 °C, and 26 °C) and colored glazing (blue, orange, and neutral) on visual perception of daylight. Experiments were performed in an office-like test room, in which 75 participants were fully immersed under visual and thermal stimuli. Findings are discussed in terms of cross-modal effects of indoor temperature on visual perception, as well as of unimodal effects of glazing color on visual perception, investigated in terms of color of light and visual environment evaluations. Results show the presence of cross-modal effects of indoor temperature on visual perception of both the visual environment and the color of light. Indoor temperature affected the visual environment evaluation as, first, brightness comfort resulted higher at lower temperatures, and second, the light was associated to warmth adjectives more often at higher temperatures. On the other hand, indoor temperature influenced the color of light evaluation as, only at lower temperatures, daylight tinted by the blue glazing was considered less pleasant and less comfortable, and was chosen less often over the daylight tinted by the orange glazing. In terms of unimodal effect of colored glazing on visual perception, results indicate that the neutral glazing was always preferred over the two colored ones, probably due to the use of saturated colors. The orange glazing, however, was associated with the most relaxing and warmest daylight. These findings have important implications regarding the use of glazing with saturated colors in buildings and expand our understanding of temperature–color interaction – so far investigated only with electric light – to daylight evaluations.

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Visual comfort; daylight; glazing; color; interactions; combined effect; indoor environment

1. Introduction

Little is known about how human visual perception is influenced by indoor factors that are not related to the visual environment, as these factors are generally either kept constant or not controlled during conventional visual comfort and perception investigations. Traditional visual perception studies, in fact, focus on the effects of factors related to the visual environment such as horizontal illuminance on the task and surrounding areas, glare, light uniformity and distribution, light quality in rendering colors, view to the outside and correlated color temperature (Aries et al. 2010; Boyce 2014; Carlucci et al. 2015; Galasiu and Veitch 2006; Wei et al. 2015, 2014). However, the interest toward the study of the effects on perception of indoor environmental factor interactions is growing (ASHRAE 2010; Bluyssen 2013; Bluyssen et al. 2018; Clements-Croome 2006; Torresin et al. 2018; Wierzbicka et al. 2018). Among all the possible combinations of factors, visual and thermal are the most commonly studied (Torresin et al. 2018). The majority of visual–thermal interaction investigations focus on the influence of visual factors on thermal responses (te Kulve et al. 2016a, 2016b). One particular example of visual–thermal interaction is the hue-heat-hypothesis (HHH), according to which color affects people's thermal perception, with bluish color leading to cold sensations and orangish color to warm sensations (Bennett and Rey 1972; Chinazzo et al. 2018a, 2019a).

In the same way, as visual factors affect the perception of the thermal environment (te Kulve et al. 2016b; Chinazzo et al. 2018a, 2019a), the *opposite* effect could occur, with occupant's visual perception varying depending on indoor thermal conditions. With reference to the HHH, the visual perception of the color of the light could therefore

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be affected by indoor thermal conditions, with a blue color preferred in warm environments and an orange color preferred in cold environments. Regardless, researchers put forward the hypothesis that the HHH could be unidirectional, meaning that the color of the light could influence thermal evaluations but that thermal stimuli could not affect the visual perception of the color of the light (Ho et al. 2014; Xu and Labroo 2014). The question is still open as a few studies exist on the specific topic (i.e., thermal factors influence on the perception of the color of the light) or more generally on the effect of thermal conditions on visual perception of light. Moreover, the majority of the existing studies were conducted in environments lit by electric light, but, as daylight and electric light have often been shown to have different effects on human perception, behavior and health (Aries 2005; Galasiu and Veitch 2006), the effect of thermal stimuli on the perception of daylight could be different from that of thermal stimuli on the perception of electric light. The only three studies investigating temperature effects on the perception of natural light did not investigate any chromatic aspects of daylight, but focused on the effect of indoor temperature on daylight quantity evaluations (Garretón et al. 2016; Chinazzo et al. 2019c; Laurentin et al. 2000).

The color variations of daylight inside buildings are related to its changing spectral power distribution (SPD) over seasons, time of the day, latitude and weather conditions, but also to the spectral properties of the glazing through which daylight is transmitted inside buildings. In particular, new glazing technologies currently installed or with the immediate potential to be used in buildings (e.g., thermochromic, electrochromic, photochromic, and gasochromic glazing types) sometimes substantially impact the spectrum of the transmitted daylight (for a detailed review and analysis of smart glazing technologies, see Aste et al. 2018). Fig. 1 shows examples of built façades that integrate spectrally selective glazing technologies with particularly saturated colors. Subjective studies conducted with some of the aforementioned smart glazing technologies have typically focused on the visual evaluation of daylight transmitted through colored glazing but have so far not really considered possible interactions with the thermal environment (Arsenault et al. 2012; Dubois 2009; Dubois et al. 2007; Liang et al. 2018). As of today, no study exists on the effect of indoor temperature on the visual perception of daylight transmitted through colored glazing.

To fill in this gap, the goal of this paper is to investigate the effect of the combined variation of indoor temperature and colored glazing on visual perception of daylight, with the use of strongly saturated colors. Variations of colored glazing imply the change of the transmitted daylight as well as that of both the appearance of the outside environment and of the objects inside the room. This research analyses only the visual perception of daylight, evaluated through subjective questionnaires. However, the authors acknowledge that perceptual responses might also be affected by the variation of the outside appearance and of the objects inside the room, considering that strongly saturated colored glazing was used in this study.

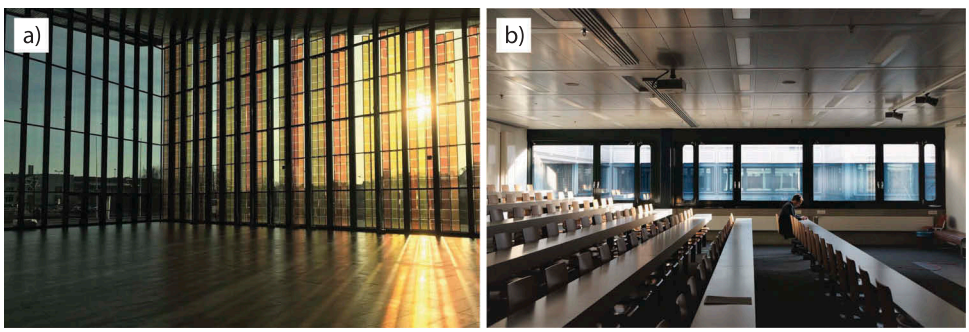


Fig. 1. Examples of existing glazing technologies with saturated colors: a) dye-sensitized solar panels in the SwissTech Convention Center at the École Polytechnique Fédérale de Lausanne (Richter Dahl Rocha & Associés. Lausanne, Switzerland); b) electrochromic glazing in an auditorium at the École Polytechnique Fédérale de Lausanne (Switzerland).

The objectives of this investigation are threefold: 1) study the effect of indoor temperature on visual perception of daylight when colored glazing is used (cross-modal *main* effect); 2) investigate whether such temperature effect is different according to color of the glazing, and/or it occurs at a particular temperature level (cross-modal *interaction* effect); 3) examine the effect of different types of colored glazing on visual perception (unimodal main effect). Visual perception is investigated by means of subjective questionnaires regarding the *color of the light* and the *visual environment* evaluations of 75 participants fully immersed under both visual and thermal stimuli in an office-like test room with controllable thermal environment and easy-to-change glazing properties.

The present study is part of a broader project investigating the interacting effects of temperature and daylight when colored glazing is used (i.e., temperature–color interaction). The general aim of the project is to understand the cross-modal effects of one factor on the other type of perception, meaning the cross-modal effect of daylight color on thermal perception (results published in Chinazzo et al. 2018a, 2019a) and the cross-modal effect of temperature on visual perception of daylight color (analyzed in the current study). In the following, after an overview of the studies on the effect of thermal conditions (as well as of glazing with a color-shift) on visual perception, we first briefly introduce the experimental design, set-up and procedure and then report results referring to the effect of temperature on visual perception. The "more traditional" effect of colored glazing on visual perception is analyzed as well. It must be remarked that, despite the presence of blue daylight in the experimental investigation, this study deals with perceptual responses linked to rod and cone-based visual perception and does not pertain to the non-image forming (NIF) effects of light, driven by non-rod and non-cone photoreceptors referred to as ipRGCs.

2. State of the art

The theoretical basis of this study finds its root in two main areas of research, categorized into two sections. The first (section 2.1) summarizes the effect of the thermal stimuli on visual perception, referring to the evaluation of the color of both object and light. Experiments on the effects of the thermal stimuli on visual perception that do not pertain to color aspects (Garretón et al. 2016; Laurentin et al. 2000; Morita et al. 2004; te Kulve et al. 2018; Yang and Moon 2018; Chinazzo et al. 2019c) were excluded from the analysis to limit the scope of the review. The second section (2.2) refers to the effect of colored light on visual perception, analyzing only investigations on daylight transmitted through glazing with a color shift (hence excluding investigations pertaining to electric lighting). Both sections refer to subjective assessments rather than to objective tasks (i.e., performance tests).

2.1. Thermal stimuli affecting visual perception of color

Significant effects of thermal stimuli on visual perception of color have been found in most of the studies in the literature, summarized in Table 1 according to the visual stimulus evaluated (either "object color" or "color of the light"). In the following, a better description of the type of visual perception investigated with reference to the two visual stimuli (e.g., color satisfaction of object color or light preference of correlated color temperature – CCT) will be provided, together with a description of the experimental stimuli and procedure, and of the results.

The four studies investigating the effect of thermal environment on *object color* evaluation (in terms of preference and satisfaction) reported all a positive effect, in line with the HHH, with bluish colors preferred in warm conditions and reddish/orangish colors in cold thermal environments (Kearney 1966; Kim and Hiromi 1998; Kim and Tokura 2001; Wang et al. 2018). The color evaluation of seven colored fabrics hanged on the wall changed according to the air temperature, with yellow and red curtains evaluated more positively at the low temperature (23 °C) and blue and green evaluated more positively in the warm environment (29 °C) (Wang et al. 2018). In another investigation manipulating the air temperature, 102 participants were asked to indicate the preferred color by means of paired comparisons (Kearney 1966). The blue hue was preferred over the red one in the warm temperature condition, whereas the red hue was preferred over the blue in the cold condition, confirming again a HHH-like result (Kearney 1966). Two peculiar

Table 1. Summary of studies investigating the effect of thermal stimuli on visual perception of color.

Authors	Visual stimulus evaluated	Thermal stimulus	Visual perception investigated	Electric lighting or daylight	Sign. effect
Wang et al. (2018)	Object color	Air temperature (static)	Color satisfaction	Electric	Yes
Kearney (1966)	Object color	Air temperature (static)	Color preference	Electric	Yes
Kim and Tokura (2001)	Object color	Face cooling	Color preference	Electric	Yes
Kim and Tokura (2001)	Object color	Body heating (hot bath)	Color preference	Electric	Yes
Nakamura and Oki (2000)	Color of the light (CCT)	Air temperature (static)	CCT preference	Electric	Yes
Nakamura and Oki (2000)	Color of the light (CCT)	Air temperature (dynamic)	CCT preference	Electric	Yes
Toftum et al. (2018)	Color of the light (CCT)	Air temperature (static)	CCT brightness	Electric	No
te Kulve et al. (2018)	Color of the light (CCT)	Air temperature (static)	CCT apparent warmth, comfort, preference, brightness	Electric	Yes

thermal stimuli were used in other two studies on object color evaluation, hot bath and face cooling (Kim and Hiromi 1998; Kim and Tokura 2001). In the first study, participants chose cooler colors after the hot bath ("body heating"), compared to choices made without the thermal stimulus (Kim and Tokura 2001). In the second study, warmer color cloths were chosen after the exposure to face cooling (air at 15 °C), compared to when participants did not receive the thermal stimulus (Kim and Hiromi 1998). Results were explained in terms of *alliesthesia* (Cabanac 1971), suggesting that object color choices were made to compensate the great shift of physiological temperature from the "*milieu interieur*" (i.e., interior medium) due to face cooling or body heating. As a general conclusion, all investigations on object color evaluations reported similar results, in line with the HHH, with bluish colors preferred in warm conditions and reddish colors in cold conditions.

The majority of the experiments evaluating the effects of the thermal environment (using air temperature, with both dynamic and static exposures) on the evaluation of the *color of the light*, focused on the evaluation of "whitish" light with changes of CCT, rather than of saturated colors (Nakamura and Oki 2000; te Kulve et al. 2018; Toftum et al. 2018). Nakamura and Oki (2000) confirmed the *alliesthesia* theory, reporting that air temperature affected the CCT preference, with warm-white light (3700 K) preferred at lower temperatures (or after a sudden decrease of temperature) and bluish light (7000 K) preferred under higher temperatures (or after a sudden increase of temperature). In another study, evaluating different aspects of the CCT of electric light, authors indicated an effect of static air temperature on several visual perception evaluations (te Kulve et al. 2018). First, in terms of apparent warmth of the light (referred to as cold-warmth association in the present study), both the warm-white (2700 K) and the cold-white light (5800 K) were perceived colder in the colder thermal environment (26 °C) compared to neutral and warm ones (29 °C and 32 °C, respectively). Then, regarding visual comfort and preference, the cold-white light (5800 K) was evaluated as the least comfortable during the cold thermal exposure, and a cooler color would have been preferred during the warm condition for both the CCT levels. Finally, in terms of brightness sensation, a higher brightness was perceived under the cold and the warm thermal conditions compared to the neutral one, for both the CCT levels (te Kulve et al. 2018). A similar trend in terms of brightness was observed in another study, investigating thermal conditions and CCT of electric light, even though temperature did not result as a significant factor from the statistical analysis (Toftum et al. 2018).

In summary, it can be concluded that thermal conditions affected visual perception evaluations of both object color and color of the light (CCT). For this reason, significant temperature effects on visual perception of daylight transmitted through colored glazing are expected as well, even though no comparable studies exist in the literature. In past investigations on the color of the light, evaluations were sought about the color of the light itself, as well as about other characteristics of

electric light such as lighting brightness and lighting apparent warmth. In the present study, the effect of indoor temperature on the two types of visual perception evaluations will be investigated, expanding the number of evaluations about other characteristics of light (from now on called "visual environment evaluations").

2.2. Glazing with a color-shift affecting visual perception

Many studies have investigated subjective perception related to colored electric light stimuli, such as those on the relationship between electric light source color and perceived brightness (Hu et al. 2006). The body of research on the effect of daylight transmitted through glazing with a color-shift on visual perception is much more limited and presents some contradictory results.

The majority of the experiments on the topic agree that a warm color-shift is preferred by occupants rather than a cold color-shift (Arsenault et al. 2012; Cuttle 1979; Liang et al. 2018; Vossen et al. 2016). Arsenault et al. (2012) indicated the bronze glazing as the most pleasant, comfortable and with the highest light level compared to blue and neutral glazing types. Their results confirmed those previously reported by other studies in terms of preference (Cuttle 1979) and light level perception (Boyce et al. 1992). The bronze window was also the preferred one in a study simulating daylight transmitted through chromatic glazing, compared to a clear and a blue condition (Liang et al. 2018). Finally, in a study evaluating the acceptability of window covering percentage of luminescent solar concentrator (LSC) characterized by a saturated red color, authors indicated that participants preferred a partially covered window (0–25% from the top) with colors in comparison to a clear glazing (Vossen et al. 2016).

On the other hand, in a study evaluating the visual perception of a four-pane window presenting a green shift, the room was perceived more enclosed, colors were evaluated drabber and daylight brightness was perceived as weaker (Bülow-Hübe 1995). The glazing with a green shift was also the most disliked one in another study, even though the effect of the color shift was confounded with that of visual transmittance (Pineault and Dubois 2008). Contrasting results are found in other studies, reporting that colored glazing had no effect on visual perception of people due to visual adaptation (Chain 2004; Cooper et al. 1973). Results might be influenced by individual and cultural differences of participants, but also by the saturation and the spatial distribution of the color used. Particularly, colors investigated did not present a strong saturation or were only partially covering the window. Additional investigations with the use of daylight and more saturated colors covering the entire glazed area are needed to expand the knowledge about the effect of colored glazing on visual perception evaluations.

2.3. Problem statement

Considering the results of past research and the lack of studies on the combined variation of indoor temperature and colored glazing (involving therefore the use of daylight), together with a research gap about the effect of saturated color glazing on visual perception, this paper aims at testing the following hypotheses:

- Indoor air temperature influences the visual perception of daylight transmitted through colored glazing, independently of the color investigated (*Cross-modal main effect*). This effect is expected in terms of the evaluation of the visual environment, rather than of color of the light. Not all the visual environment evaluations considered in this study were collected in past investigations. However, according to the results previously described, it is expected that daylight is perceived as brighter in a thermally uncomfortable environment and that the apparent warmth of light increases with temperature independently of the color of the glazing used.
- Indoor air temperature influences the visual perception of daylight transmitted through colored glazing, but results depend on the level of temperature and on the color (*Cross-modal interaction effect*). This effect is expected in terms of color of the light evaluations. More specifically, an interaction between temperature and daylight transmitted through colored

glazing is expected, meaning that the evaluation of the color of the light depends on the level of indoor temperature and on the color evaluated: it is foreseen that orange daylight is preferred and more comfortable at lower temperatures and that blue daylight is preferred and more comfortable at higher temperatures.
- Colored glazing affects the visual perception evaluation, with a strong orange color covering the entire surface preferred than a blue and the neutral one (*Unimodal effect*).

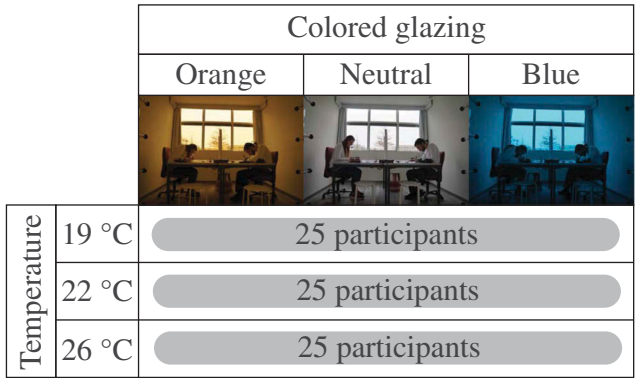


Fig. 2. Experimental design showing the three levels for each of the two factors of the design. The order of the colored daylight levels was counterbalanced across participants and temperature levels.

3. Method

3.1. Participants

A total of 75 participants were recruited within undergraduate students in the University through an e-mail invitation. To avoid having a biased sample, the invitation was not sent to students following courses taught by the researchers' laboratory or that generally studied disciplines related to the investigation (i.e., architecture as well as civil and environmental engineering students were excluded). For the same reason, people working in the investigators' laboratory were not included. Gender balance was respected across the temperature levels investigated (experienced by a total of 25 participants each), with 10 females and 15 males experiencing each temperature level. The pre-selection process, based on an online questionnaire, allowed to select participants between 18 and 30 years old, that did not assume drugs and did not abuse of alcohol, with full color vision and in good health, which were French speaking (mother tongue or at least C2 level) and with a BMI (i.e., Building Mass Index) in between 18 and 25 kg/m^2. Participants were paid for taking part in the study and provided written informed consent prior to the beginning of the experiment.

3.2. Design

A 3 × 3 full factorial design was used to investigate the combined effect of colored glazing and temperature on the perception of the visual environment. Fig. 2 illustrates the three levels of colored glazing (orange, neutral, and blue) and the three levels of temperature (19 °C, 22 °C, and 26 °C). The temperature levels were chosen to have a comfortable thermal environment (22 °C) and two slightly uncomfortable ones (19 °C slightly cold and 26 °C slightly warm) compared to temperature ranges usually found in buildings. However, the reported thermal sensation (results illustrated in the complementary study, Chinazzo et al. 2018a) resulted slightly colder for all three temperature levels, with the 26 °C associated to a "neutral" thermal sensation rather than to a "slightly warm" as expected. The implications of these reported thermal sensations will be analyzed in the Discussion section.

The study followed a mixed-design with one session for each participant. The colored glazing was the within-subject factor, while the temperature was the between-subjects factor. Consequently, participants were exposed for the entire duration of the experiment to one temperature only, while they experienced all three colored glazing levels, presented in a counterbalanced order across participants and temperature levels. Experimental sessions, planned in both morning and afternoon (with temperature levels randomized and not counterbalanced across the two times of the day), started at the same time of the day (starting time ranged from 8:30 to 9:00 or from 13:30 to 14:00) and were not performed in case of rain or partially cloudy sky to avoid large variations of illuminance values. Due to weather conditions and to participants' availability, 83% of the experiments were run in the morning and 17% in the afternoon.

3.3. Experimental set-up

The experiment was performed in an office-like test room with controllable radiant temperature on all the opaque surfaces and daylight entering from the north and the south façade. The test room, 6.55 m deep, 3.05 m wide, and 2.65 m high, was able to host the researcher and two participants at a time, occupying two facing desks close to the north window (to avoid direct sun on the working plane and the risk of glare, although direct sunlight entered the south window during sunny days). The north window was facing an open space with green areas and a parking lot, surrounded by buildings. The level of visual and acoustic disturbance from the parking lot was very low as the great majority of cars arrived in the early morning and left in the late afternoon, outside the experimental hours. Figure 3 shows the north opening of the test room (and the view to the outside open space), with the two desks of the participants. Besides the latter, the test room had an additional place for the researcher close to the south façade and was equipped with instruments for the continuous recording of thermal and visual indoor environmental parameters. In particular, both air and globe temperatures were measured at roughly 50 cm from each participant at four heights (0.1 m, 0.6 m, 1.1 m, and 1.6 m according to standard EN ISO 7726:2002). The air temperature sensors were protected from radiation by a cylindrical reflective metal cover and the globe temperature sensors had 7 cm of diameter and a black color. In addition to this, the surface temperature of the wall behind each participant was measured by iButtons® data loggers (DS1921H-F5, Maxim Integrated), wireless sensors attached to the walls. Glazing color was changed with the application of foil-filters with similar visible transmittance (~40%) attached to a PMMA transparent support, placed on the existing window frames. The SPD of the transmitted daylight was recorded at a 1-min sampling rate with spectrometers located close to participants (between them and the north window) equipped with a cosine corrector, and measuring the spectrum of the incoming daylight at the eye level of participants in a direction parallel to the window (the supposedly primary view direction of participants considering the task they performed). The measured SPDs with this configuration, considering the direction, location and presence of the cosine corrector, were considered a good approximation of what experienced by participants.

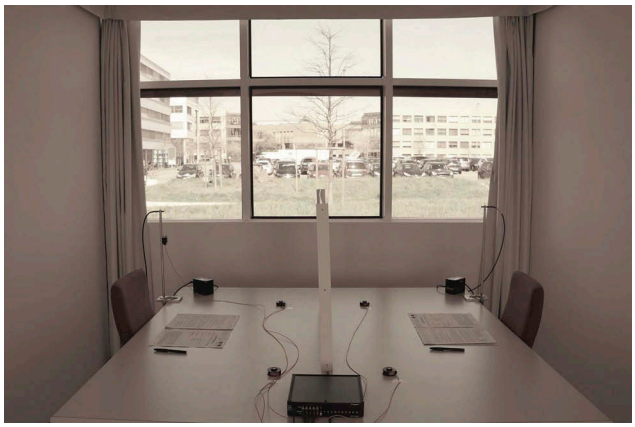


Fig. 3. Experimental setting showing the north façade of the test room and the two desks of participants.

3.4. Procedure

Two participants at a time took part in each experimental session, lasting in a total of 3 h. Fig. 4 illustrates the procedure. The pretest period of 45 min, consisted in the explanation of the experiment following a single-blind procedure (to avoid biasing participants' responses), signature of information and consent sheet, explanation of the office tasks to be performed during the experiment (paper-based task; results not reported in this paper but discussed in Chinazzo et al. 2019b) and a questionnaire about personal data. During this period, electric light was turned on (at 400 lux at the desk level, 3800 K) and daylight was blocked from entering the room by blackout curtains. The pretest period was necessary to allow thermal and metabolic adaptation of participants, guaranteeing a similar starting point for the successive color exposures.

In each of the three colored glazing levels, participants followed the same procedure, which lasted approximately 30 min. Before each color exposure, a 10-min break allowed to change the color of the glazing of the room, with the application of colored filters. During this time, participants were blindfolded and were asked to relax. At the end of each colored exposure, participants

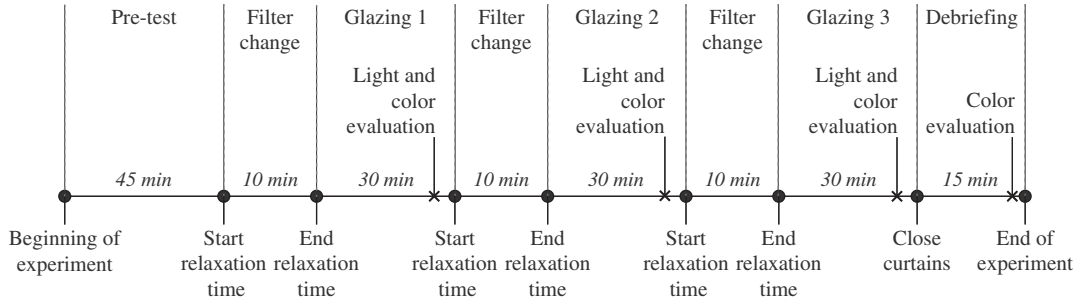


Fig. 4. Experimental procedure.

filled in a questionnaire to express their evaluation of the visual environment inside the test room. A debriefing period of 15 min concluded the experiment. At the end of it, participants evaluated the three colored glazing experienced in the previous hours.

The study took place from November 2016 to May 2017 after approval by the Ethics Committee (ref. No. 2016–01115), and was conducted in accordance with the ethical standards in the 1964 Declaration of Helsinki.

3.5. Questionnaire

Visual perception evaluations were collected with a questionnaire presented on a tablet (180 x 262 × 8.9 mm, PLS LCD display with a White LED backlight, high-resolution 1280 × 800 pixel display with a screen Aspect Ratio of 1.60). The questionnaire was designed based on factors evaluated in studies on color and light, such as comfort, sensation, and preference (Arsenault et al. 2012; Shamsul et al. 2013; te Kulve et al. 2016a) and adapting standardized questions of thermal perception, investigated in the same experiment (results reported in Chinazzo et al. 2018a).

Studies analyzed in the literature review in section 2.1 were categorized according to the visual stimulus investigated, namely object color and color of the light. In this study, we do not focus on object color inside the room (nor on the evaluation of the appearance of the outside view), but we only analyze responses related to the transmitted daylight (even though, as already anticipated in the introduction, we acknowledge that responses regarding daylight might be affected by any information available to participants, including the color of objects in the room and the appearance of the outside view). Regarding daylight evaluation, participants were asked to evaluate both the *color of the light* by directing their attention toward potential color shifts, and the *visual environment*, with different visual perception ratings. Tables 2 and 3 report the questions for the two types of evaluation. Some of the questions regarding the visual environment can be further subcategorized in terms of *light quantity evaluation* and the *light quality evaluation*, as indicated in Table 3. These terms will be used in the result and discussion sections. A unique code defines the evaluation category (CL = Color of the Light; VE = Visual Environment) and the time of presentation of the question, either during the color exposure

Table 2. Color of light evaluation questions (translated from French).

Question	Response scale or item
CL_DE1. Color comfort: How do you find the color of the light in this room?	Very comfortable (5), comfortable (4), slightly uncomfortable (3), uncomfortable (2), very uncomfortable (1)
CL_DE2. Color acceptability: How do you judge the color of the light in this room?	Unacceptable (0) – Acceptable (100)
CL_AE1. Color evaluation: Imagine that you are in the room to work. Assign a grade to each colored glazing from 1 to 10, with 1 being not at all pleasant and 10 very pleasant.	Orange: _____ Blue: _____ Neutral: _____
***CL_AE2. Color ranking:** Rank the colors of the glazing that you have seen during the experiment according to your preference.	First: _____ Second: _____ Third: _____

* Responses are grouped into two categories, one gathering responses for the first and second choice and the other for the third choice.

Table 3. Visual environment evaluation questions (translated from French).

	Question	Response scale or item
Light *quantity* evaluation	**VE_DE1. Brightness sensation**: How do you perceive the quantity of light on your desk?	Very low (1), low (2), slightly low (3), adequate (4), slightly high (5), high (6), very high (7)
	VE_DE2. Brightness comfort: How do you find the quantity of light on your desk?	Very comfortable (5), comfortable (4), slightly uncomfortable (3), uncomfortable (2), very uncomfortable (1)
	VE_DE3. Brightness preference: In this moment, would you like to have a light on your desk:	Much brighter (1), brighter (2), slightly brighter (3), no change (4), slightly dimmer (5), dimmer (6), much dimmer (7)
	VE_DE4. Overall visual environment comfort: How do you find the general visual environment in the room?	Very comfortable (5), comfortable (4), slightly uncomfortable (3), uncomfortable (2), very uncomfortable (1)
	***VE_DE5. Discomfort reason**: You have said that you are not comfortable with the general visual environment. Which of the following has contributed to your discomfort?	Color of the glazing, unpleasant exterior view, blinds not operable, lack of electrical lighting, glare from the window, window too far, window too close, window too big, window too small,, environment too dim, environment too bright, reflections on the screen, too much direct light, not enough direct light, light distribution not uniform, other
Light *quality* evaluation	**VE_DE6. Lighting cold-warmth association**: Please specify how do you perceive the Lighting in the indoor environment in this moment:	Cold (1) – Warm (7)
	VE_DE7. Lighting comfort: Please specify how do you perceive the light in the indoor environment in this moment:	Uncomfortable (1) – Comfortable (7)
	VE_DE8. Lighting pleasantness: Please specify how do you perceive the light in the indoor environment in this moment:	Unpleasant (1) – Pleasant (7)
	VE_DE9. Lighting relaxation: Please specify how do you perceive the light in the indoor environment in this moment:	Stimulating (1) – Relaxing (7)

* This question appeared only when participants indicated to be slightly uncomfortable, uncomfortable, or very uncomfortable in question VE_DE4. In the analysis, responses are grouped into "unpleasant color of the glazing" and "others."

(DE = During Exposure) or during the debriefing phase, after the presentation of the three colored glazing (AE = After Exposure). Each question is also given a title, which is used in the result and discussion sections.

3.6. Statistical analysis

The statistical analysis was performed in two parts. In the first part, which represents the main analysis, linear or generalized linear mixed-effects model analyses were conducted on the full dataset to investigate the main effect of colored glazing, temperature, and their interaction on visual perception evaluations. This type of analysis was chosen as, to the authors' knowledge, it is the only statistical method adequate to analyze interactions with results coming from a mixed-effects model design. In this type of analysis, the repeated-measures for each colored glazing were taken into account by modeling a random intercept for each participant, using their identification code as random factors of the model (Pinheiro and Bates 2000). Linear mixed-effects model analyses were conducted for all the questions, except for the color ranking (CL_AE2), as it was post-processed as a binary categorical response and required to be analyzed with a logistic regression (generalized linear mixed-effects model, binomial family).

For each of the questions, the fixed effects modeled were the glazing color and the temperature, both treated as categorical values to consider the three levels within each of the two factors (i.e., neutral, orange, and blue for the colored glazing factor and 19 °C, 22 °C, and 26 °C for the temperature factor). The interaction between the two factors (Color x Temperature) was also considered as a fixed factor, and was excluded from the model when it was shown not to be significant. Whenever the color of the glazing or the temperature resulted as a significant factor, post-hoc pairwise comparisons for all the possible combinations of the levels of the factors were conducted applying the multiplicity adjustment Tukey's HSD to control for false discovery rate (Coolican 2014). This was done to compare results across color levels whenever glazing color emerged as a significant factor (with results collapsed across temperatures) or to compare results across

temperature levels whenever the temperature factor was significant (considering results collapsed across color levels). Few additional covariates were included in the analysis, representing factors that could have influenced visual evaluations as well, and that were not fully controlled during the experiments: measured illuminance level at the desk for each participant, morning or afternoon experimental session, sky condition (sunny or overcast), sex, and outdoor running mean temperature (RMOT) calculated according to the standard EN ISO 15251 (2008). The RMOT was used as a covariate instead of considering the season in which the experimental sessions were performed (according to equinoxes and solstices) due to unusual outdoor temperature values recorded in 2016 and 2017.

The second part of the statistical analysis was conducted in case of significant interaction effect or whenever the exploratory data analysis, performed by plotting the data, revealed a possible interaction between temperature and color (although not detected by the statistical analysis), by showing differences between only two colored glazing levels and/or at only one temperature level. This additional analysis was performed on the full dataset or on the reduced one (considering only two colored glazing levels) at each temperature level with mixed-effects model analysis to consider for the repeated measurements at each color level. In case of a significant effect of colored glazing, post-hoc pairwise comparisons across color levels were performed.

We recognize the importance of reporting effect sizes in the statistical analysis whenever possible (e.g., as done in another publication by the authors in which another statistical test was used – Chinazzo et al. 2018b). However, we were not able to calculate the effect sizes of our results as, to the best of our knowledge, it is still not possible to estimate the effect size of mixed-effects models with more than one fixed effects as in the present study (we acknowledge however the possibility to perform the calculation of the effect size for simple designs with one fixed effect and two random factors as reported in Brysbaert and Stevens 2018).

Analyses were conducted using the software R (R Core Team 2017) and the packages "lmerTest" (Kuznetsova et al. 2017) and "car" (Fox and Weisberg 2011). The additional packages "lsmeans" (Lenth 2016) and "multcomp" (Hothorn et al. 2008) were used to conduct post hoc pairwise comparisons between the levels of temperature or colored glazing whenever one or both of these factors were significant. All graphs were created with the "ggplot2" package (Wickham 2009). The significance level for all analyses was set at 0.05.

4. Results

After a description of the measured indoor environmental conditions, results are discussed first in terms of color of the light (section 4.2), then of visual environment (section 4.3). Within each of the two main sections, results are divided according to the effects of colored glazing, of temperature (and its interaction with colored glazing), and of confounding factors.

4.1. Indoor environmental conditions

Indoor environmental factors were continuously measured during the experimental sessions. The mean indoor temperature did not differ between color exposures over time. Fig. 5 reports the average values of all the spectral power distribution measurements (recorded at 1-min sampling rate) resulting from the three colored glazing types, plotted relative to the peak of the blue glazing. CIE chromaticity coordinates were calculated for all the measurements as well (average values: blue $x = 0.20$, $y = 0.30$; orange $x = 0.55$, $y = 0.41$; neutral $x = 0.31$, $y = 0.33$) and did not change according to time of the day or season. Although experimental days were selected to have stable weather conditions, illuminance values were not constant during the experiments. Nevertheless, their variation was limited, and their values were considered to be in comfortable ranges (Table 4). In any case, as already indicated, measured illuminance values were considered in the statistical analysis to take into consideration their variation.

4.2. Color of light evaluation

Table 5 reports the results of the first part of the statistical analysis, conducted on the full dataset. It includes significant main and interaction effects

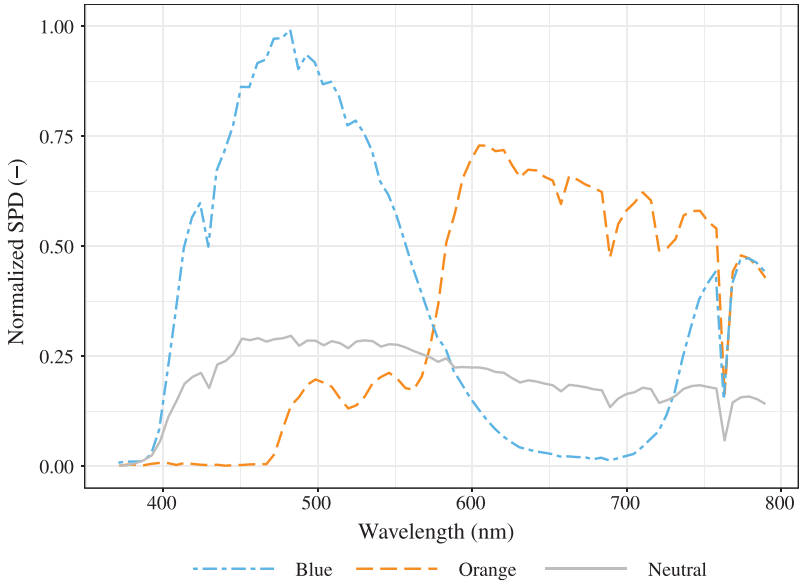


Fig. 5. Average spectral power distribution (SPD) of daylight transmitted through each of the three glazing.

Table 4. Measured horizontal illuminance values in each colored glazing-temperature combination (average over the entire duration of each condition ± standard deviation).

	Colored glazing		
Temperature	Blue	Neutral	Orange
19 °C	618 ± 163	605 ± 150	474 ± 133
22 °C	565 ± 194	475 ± 172	419 ± 178
26 °C	645 ± 178	565 ± 124	509 ± 102

on the color of light (CL) evaluations, together with results of the post-hoc tests whenever main effects are significant (non-significant effects are not reported but they are provided as supplementary material). Results of the second part of the statistical analysis and the motivation for its implementation are described in the text. In the following, results are divided according to the main effects of colored glazing, main and interaction effects of temperature, and the effects of the confounding factors on CL evaluations.

4.2.1. Main effects of colored glazing on CL evaluations

As expected, colored glazing affected all the evaluations of the color of light (Table 5), resulting in a significant factor for color comfort (CL_DE1, Fig. 6a), acceptability (CL_DE2, Fig. 6b), evaluation (CL_AE1, Fig. 6c) and ranking (CL_AE2, Fig. 6d). Post-hoc comparisons indicated a significant difference between neutral and the two colored glazing for all the color of the light evaluations, as well as between orange and blue for the color ranking rating

Table 5. Results from the mixed-effects model analysis for the full dataset (reporting only significant effects) and multiple comparisons (significant results in italics) for the color of the light evaluations. The *p*-values are corrected with the multiplicity adjustment Tukey's HSD. Significant effect with "*" $p < .05$, "**" $p < .01$, "***" $p < .001$.

Question	Significant effect	Mixed-effects model result	Pairwise comparison	Mean difference	Std. Error	df	t ratio (z value†)	p value
Color comfort [CL_DE1]	Colored glazing	$F(2,215) = 28.69$, $p < .001$***	*Blue – Neutral*	−1.06	0.14	219	−7.47	*<.0001****
			Blue – Orange	−0.23	0.14	219	−1.64	.22
			Neutral – Orange	0.82	0.14	219	5.83	*<.0001****
Color acceptability [CL_DE2]	Colored glazing	$F(2,215) = 39.95$, $p < .001$***	*Blue – Neutral*	−28.05	3.45	222	−8.11	*<.0001****
			Blue – Orange	−2.50	3.45	222	−0.72	.75
			Neutral – Orange	25.54	3.45	222	7.39	*<.0001****
Color evaluation [CL_AE1]	Colored glazing	$F(2,215) = 64.28$, $p < .001$***	*Blue – Neutral*	−3.89	0.36	222	−10.74	*<.0001****
			Blue – Orange	−.068	0.36	222	−1.87	.14
			Neutral – Orange	3.21	0.36	222	8.86	*<.0001****
Color ranking [CL_AE2]	Colored glazing	$F(2,222) = 16.14$, $p < .001$***	*Blue – Neutral*	3.17	0.56	222	5.62†	*<.001****
			Blue – Orange	0.81	0.33	222	2.43†	*.037**
			Neutral – Orange	−2.35	0.56	222	−4.16†	*<.001****

(CL_AE2, Table 5). In particular, the neutral glazing resulted as the most comfortable, acceptable, and pleasant option compared to the orange and the blue one, besides being the most selected first choice when a ranking was asked to participants. Except for color ranking, responses between the blue and the orange glazing did not differ when the whole sample was analyzed for color comfort, acceptability, and pleasantness (first part of the analysis). However, from a visual inspection of the data (Fig. 6), it appeared that responses between the blue and the orange evaluations differed only in the cold thermal condition, indicating that a possible interaction between temperature and the color could have occurred, possibly for all the four color of light evaluations. Further evaluation of this interaction through additional analyses (second part of the statistical analysis) are reported in the next section.

4.2.2. Main and interaction effects of temperature on CL evaluations

For the four color of light evaluations, temperature, and the interaction term between

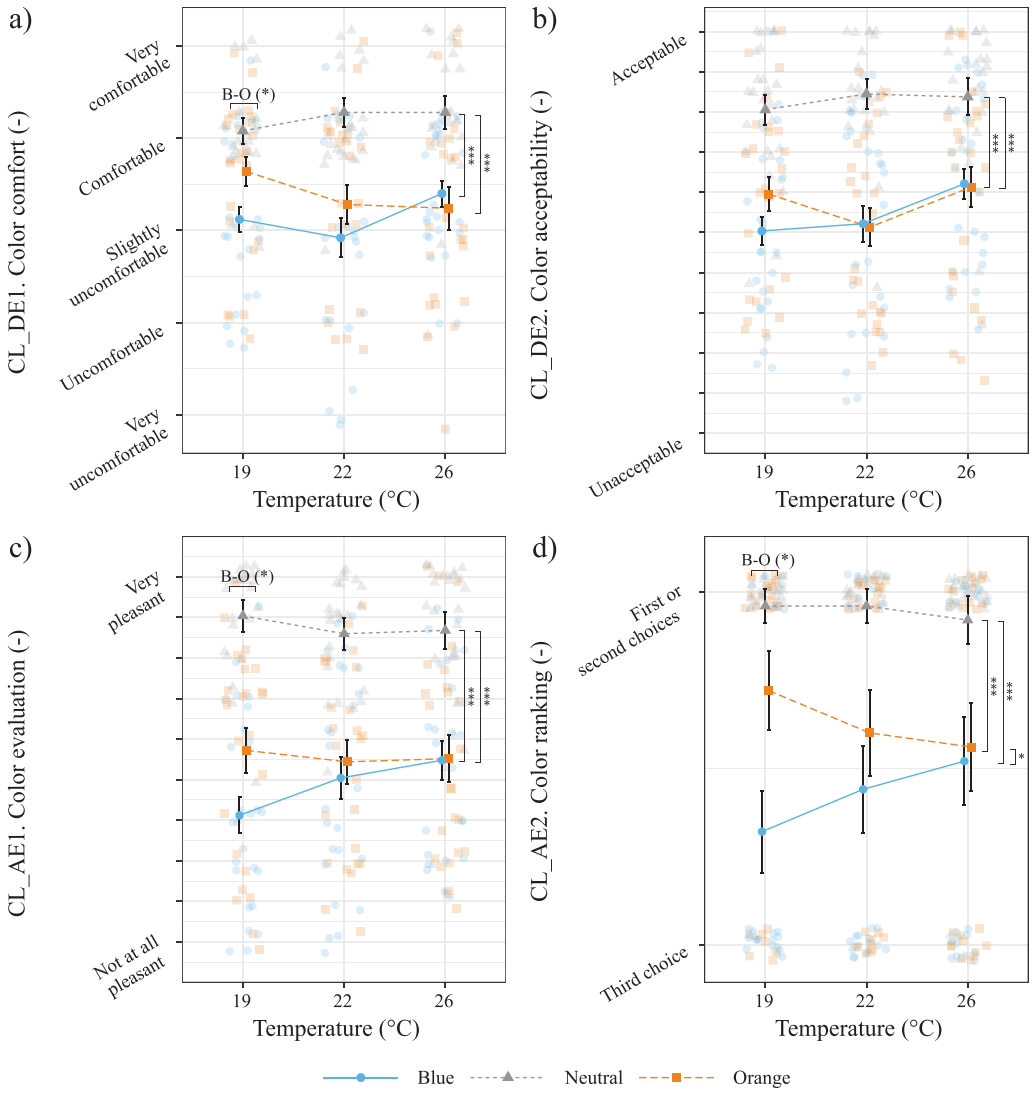


Fig. 6. Average and s.e.m. visual perception evaluations about the color of the light for each colored glazing and temperature level combination. a) Color comfort; b) color acceptability; c) color evaluation and d) color ranking. The transparent points in the background represent the raw data. Main effect of colored glazing indicated on the side with vertical lines, main effect of temperature indicated on the top with horizontal lines and significant difference between two colored glazing (B = blue, O = orange) at a particular temperature in case of significant interactions. Significance levels indicated with "*" $p < .05$, "**" $p < .01$, "***" $p < .001$.

temperature and colored glazing were never significant factors when the analysis was performed with the full dataset in the first part of the statistical analysis. However, as indicated in the previous section, additional data evaluations were conducted at each temperature level as the visual inspection of the data suggested possible interactions between temperature levels and the blue and the orange glazing (Fig. 6). Results of this second part of the statistical analysis, conducted considering only the evaluations with the blue and the orange glazing, show that significant differences between the orange and the blue daylight were present, but only at the low-temperature level (19 °C). This occurred for color comfort ($CL_DE1, blue - orange = -0.49, p = 0.041$), evaluation ($CL_AE1, blue - orange = -1.60, p = 0.031$), and ranking ($\text{CL_AE2}, \text{blue} - \text{orange} = 1.69, p = 0.015$). In particular, only at 19 °C, the orange glazing was always better rated compared to the blue one (Fig. 6a,c,d). With increasing temperatures, the evaluation of the blue glazing increased and that of the orange one decreased (or remained stable), leading to a similar evaluation under both the blue and the orange glazing at the higher temperature tested (26 °C). This result indicates an interaction between temperature and the blue and the orange glazing (cross-modal interaction effect). The interaction term was not significant when the full dataset was analyzed as differences between blue and orange in two out of three temperature levels (i.e., 22 °C and 26 °C) were not significant. We hypothesize that an opposite result (blue more comfortable, more pleasant and preferred over orange) would (potentially) appear at higher temperature levels than those we investigated in our study. It is worth noticing that people exposed to 26 °C, on average, rated the environment as "neutral" on the thermal sensation scale (Chinazzo et al. 2018a). We assume that a "slightly warm" or "warm" thermal environment could therefore lead to higher evaluations of the blue glazing over the orange one and hence to a stronger temperature–color interaction, detectable for the full dataset.

4.2.3. Effects of confounding factors on CL evaluations

Sex, time of the day, illuminance level, sky condition, and outdoor temperature were never significant factors for the four colors of the light evaluations.

4.3. Visual environment evaluation

Table 6 reports the results of the first part of the statistical analysis on the nine questions about the visual environment (VE) evaluation, highlighting only significant main and interaction effects of temperature and colored glazing (non-significant effects are not reported but they are provided as supplementary material), together with results of the post-hoc tests. In the following, the results of all questions are discussed, whereas only relevant results are plotted in Fig. 7. Results are divided according to the main effects of colored glazing, main and interaction effects of temperature, and the effects of the confounding factors.

4.3.1. Main effects of colored glazing on VE evaluations

Colored glazing resulted as a significant factor for all questions referring to the visual environment, in terms of light *quantity* evaluation (brightness sensation – VE_DE1; brightness comfort – VE_DE2; brightness preference – VE_DE3), overall visual environment comfort (VE_DE4) and light *quality* evaluation (lighting cold-warmth association – VE_DE6; lighting comfort – VE_DE7; lighting pleasantness – VE_DE8; lighting relaxation – VE_DE9).

In terms of light *quantity* evaluation, contrary to responses on the color of the light (section 4.2.1), results were not always determined by the differences between the neutral and the other two colors, as shown by the post-hoc comparisons (Table 6). Daylight brightness (brightness sensation – VE_DE1) was significantly lower for the orange glazing compared to the blue and the neutral ones, and a significantly brighter environment (brightness preference VE_DE3) was preferred under orange compared to neutral, but not to blue. On the other hand, brightness comfort (VE_DE2) resulted higher for the neutral glazing in comparison to both the orange and the blue ones (Fig. 7a). It must be noted, however, that the

Table 6. Results from the mixed-model analysis for the full dataset (reporting only significant effects) and multiple comparisons (significant results in italics) for visual environment evaluations. The *p*-values are corrected with the multiplicity adjustment Tukey's HSD. Significant effect with "*" $p < .05$, "**" $p < .01$, "***" $p < .001$.

Question	Significant effect	Mixed-effects model result	Pairwise comparison	Mean difference	Std. Error	df	t ratio	p value
Brightness sensation [VE_DE1]	Colored glazing	F(2,215) = 9.66, $p = .001$***	Blue – Neutral	−0.22	0.11	222	−2.09	.09
			Blue – Orange	0.30	0.11	222	2.82	*.01***
			Neutral – Orange	0.53	0.11	222	4.91	*<.0001****
Brightness comfort [VE_DE2]	Colored glazing	F(2,215) = 22.81, $p < .001$***	*Blue – Neutral*	−0.64	0.11	219	−5.82	*<.0001****
			Blue – Orange	0.005	0.11	219	0.047	.99
			Neutral – Orange	0.65	0.11	219	5.87	*<.0001****
	Temperature	F(2,215) = 7.46, $p = .007$**						
Brightness preference [VE_DE3]	Colored glazing	F(2,215) = 7.53, $p < .001$***	Blue – Neutral	−0.25	0.10	222	−2.34	.053
			Blue – Orange	0.2	0.10	222	1.84	.15
			Neutral – Orange	0.45	0.10	222	4.19	*.0001****
Overall visual environment comfort [VE_DE4]	Colored glazing	F(2,215) = 12.65, $p < .001$***	*Blue – Neutral*	−0.52	0.11	222	−4.66	*<.0001****
			Blue – Orange	−0.08	0.11	222	−0.71	.75
			Neutral – Orange	0.44	0.11	222	3.95	*.0004****
Discomfort reason [VE_DE5]	Colored glazing x temperature	F(4,215) = 2.50, $p = .04$*						
Lighting cold-warmth association [VE_DE6]	Colored glazing	F(2,215) = 93.56, $p < .001$***	*Blue – Neutral*	−1.16	0.18	221	−6.19	*<.0001****
			Blue – Orange	−2.78	0.18	221	−14.87	*<.0001****
			Neutral – Orange	−1.62	0.18	221	−8.68	*<.0001****
	Temperature	F(2,215) = 6.46, $p = .002$**						
Lighting comfort [VE_DE7]	Colored glazing	F(2,215) = 16.93, $p < .001$***	*Blue – Neutral*	−1.24	0.25	222	−4.94	*<.0001****
			Blue – Orange	0.05	0.25	222	0.21	.97
			Neutral – Orange	1.29	0.25	222	5.15	*<.0001****
Lighting pleasantness [VE_DE8]	Colored glazing	F(2,215) = 15.97, $p < .001$***	*Blue – Neutral*	−1.26	0.24	222	−5.29	*<.0001****
			Blue – Orange	−0.18	0.24	222	−0.78	.71
			Neutral – Orange	1.08	0.24	222	4.51	*<.0001****
Lighting relaxation [VE_DE9]	Colored glazing	F(2,215) = 14.04, $p < .001$***	Blue – Neutral	−0.18	0.21	222	−0.87	.65
			Blue – Orange	−1.25	0.21	222	−5.87	*<.0001****
			Neutral – Orange	−1.06	0.21	222	−5.00	*<.0001****

experiment was not designed specifically for the evaluation of the light quantity and that illuminance variations within experimental sessions could have confounded the results. Nevertheless, illuminance values varied within comfortable ranges and the statistical analysis did not show any effect of the factor "illuminance" considered as a covariate. To confirm the obtained results, however, specific investigations should further be performed with a focus on light quantity evaluations, varying both color and illuminance in a controlled manner.

A similar trend as for the color of light evaluations (i.e., neutral more positively evaluated than the other two colored glazing types) was observed for overall visual environment comfort (VE_DE4), and for two questions of the light *quality* evaluation – i.e., lighting comfort (VE_DE7) and lighting pleasantness (VE_DE8). The fact that four questions considered similar as they evaluate visual comfort and pleasantness (i.e., brightness comfort – VE_DE2, overall visual environment comfort – VE_DE4, lighting comfort – VE_DE7, and lighting pleasantness – VE_DE8), report the same conclusions, can be considered as an internal validation of the questionnaire. Responses for lighting cold-warmth association (VE_DE6) were significantly different between all three levels of colors, with orange color indicated as the warmest and the blue as the coolest. Orange daylight was also considered the most relaxing visual condition compared to both the neutral and the blue ones (Fig. 7c).

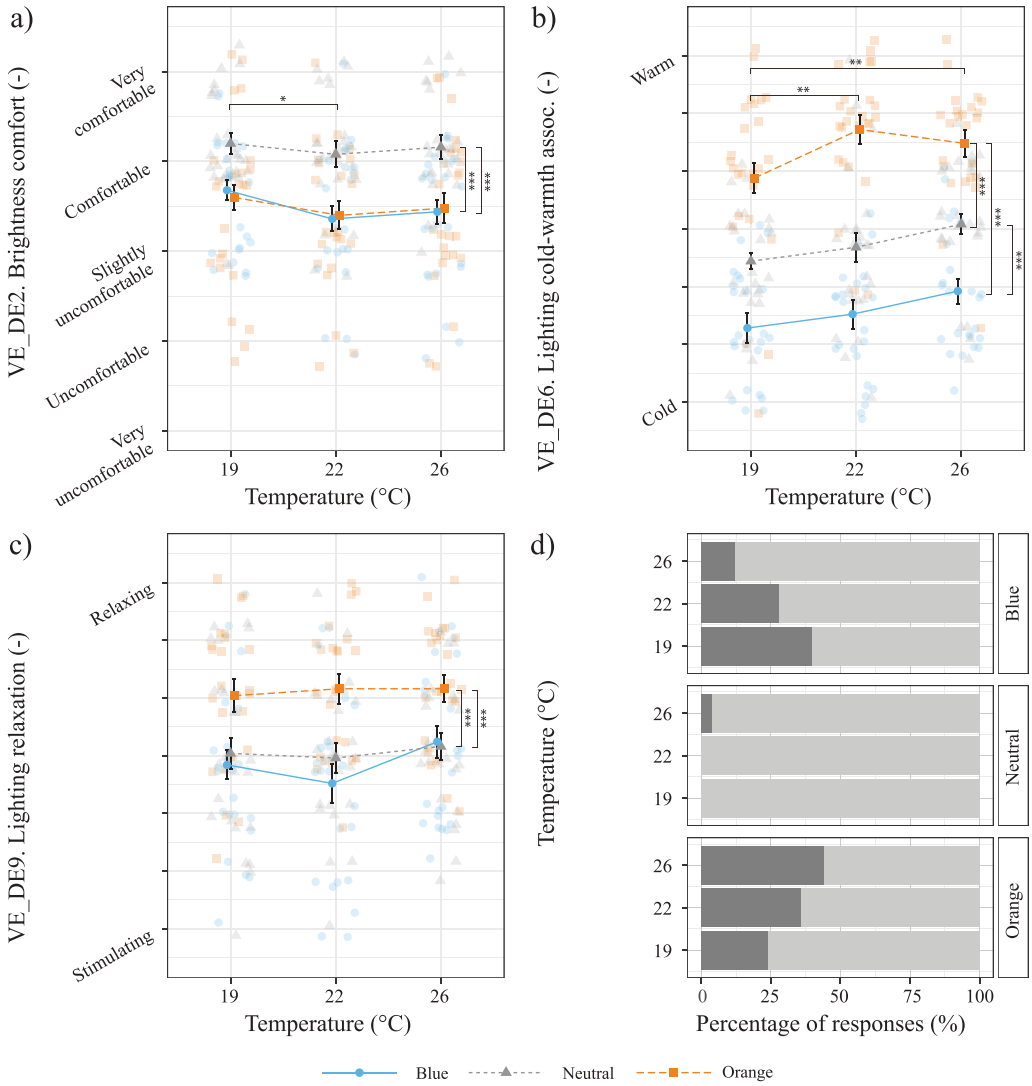


Fig. 7. Average and s.e.m. visual perception evaluations about visual environment for each colored glazing and temperature level combination. a) Visual comfort; b) lighting cold-warmth association and c) lighting relaxation. The transparent points in the background represent the raw data. d) Percentage of people indicating the uncomfortable color of the glazing as the reason of visual discomfort for each temperature and colored glazing combination. Main effect of colored glazing indicated on the side with vertical lines, main effect of temperature indicated on the top with horizontal lines. Significance levels indicated with "*" $p < .05$, "**" $p < .01$, "***" $p < .001$.

4.3.2. Main and interaction effects of temperature on VE evaluations

When the analysis was conducted with the full dataset in the first part of the statistical analysis, temperature was not a significant factor for the majority of the questions on the visual evaluation, except for brightness comfort (VE_DE2) and lighting cold-warmth association (VE_DE6) (Table 6). In particular, brightness comfort was higher for all the colored glazing levels at the low-temperature level (19 °C) compared to the neutral one (22 °C) (Fig. 7a), indicating a cross-modal main effect of indoor temperature on such visual perception evaluation. Then, participants associated the light of all colored glazing types to a warm adjective more often at the higher temperatures (22 °C and 26 °C) compared to the lower one (19 °C) (Fig. 7b), indicating again a cross-modal main effect of indoor temperature on the visual environment evaluation. Such temperature effect on the two evaluations of the visual environment was independent of color as the same effect occurred under the three colored glazing and the interaction term was not significant.

The interaction term was a significant factor for only one visual evaluation question: the reason of visual discomfort (Table 6, VE_DE5). Participants complained the most about uncomfortable color of the glazing under blue color at the low temperature, and under orange color at high temperature (Fig. 7d).

4.3.3. Effects of confounding factors on VE evaluations

Sex, time of the day, illuminance level, sky condition, and outdoor temperature were not significant factors for the nine questions on the visual environment evaluation.

5. Discussion

The present study investigated the combined effect of indoor temperature and colored glazing on visual perception, with a focus on the transmitted daylight, evaluated in terms of the color of the light and the visual environment. Three hypotheses were proposed in terms of cross-modal *main* effect of indoor temperature, cross-modal *interaction* effect of indoor temperature and colored glazing, and unimodal effect of colored glazing on visual perception. The following discussion will be divided according to the aforementioned hypotheses and will be followed by the study limitations. Due to the lack of studies on the specific focus of this investigation, comparisons with previous literature are possible only by considering studies on electric light or on glazing with a slight color-shift (or with saturated glazing covering only part of the glazed opening, as reported in the state of the art analysis).

5.1. Cross-modal main effects of temperature on visual perception

In line with the findings of previous studies on electric light (Table 1), our results indicate that visual perception of daylight was affected by indoor temperature, independently of the colored glazing (cross-modal *main* effects). How expected, this occurred only in terms of visual environment evaluation and not for the color of the light evaluation (for which cross-modal *interaction* effects occurred, discussed in the next section). However, not all the visual environment responses were affected by the indoor temperature, and responses were not always in line with results of past investigations: only brightness comfort and lighting cold-warmth association varied according to the thermal environment.

In terms of brightness comfort, the lowest temperature level (19 °C) resulted in a higher brightness comfort score than that reported for the higher temperature (22 °C). We speculate that the increased brightness comfort might be due to a negative correlation with the thermal comfort, which decreased at lower temperatures, as a negative evaluation of one indoor environment aspect (i.e., thermal) might result in a higher forgiveness of another, secondary aspect (i.e., visual). This interpretation is in contrast with the one presented in another study, where a positive correlation between visual and thermal comfort was indicated, but only in a cold and warm thermal environment (te Kulve et al. 2018). As the effect of indoor temperature on brightness comfort is stronger for the two colored glazing types (orange and blue) compared for the neutral color glazing (showing a quite constant evaluation at all indoor temperature levels), we hypothesize that the negative correlation between visual and thermal comfort might be due to the use of saturated colors, evaluated less pleasant than the neutral one. In other words, thermal discomfort might have compensated visual discomfort of the two visually uncomfortable saturated glazing types, resulting in a higher brightness comfort in the low thermal condition. Further investigations would be necessary to confirm this initial result.

Regarding the lighting cold-warmth association, apparent warmth increased with temperatures as the daylight resulting from the three colored glazing types was associated more often to a warm adjective at warmer temperature levels (22 °C and 26 °C) compared to at the lowest one (19 °C). This result was independent of the colored glazing level and agrees with the one reported in another study on electric light with different CCTs (te Kulve et al. 2018). As the visual perception changed according to the temperature level (besides its dependence on the color of the glazing, as discussed in section 5.3), this result indicates that people's perception of a specific environmental aspect is unconsciously affected by other indoor factors.

Contrary to past investigations where significant effects of indoor temperature on brightness perception were found (te Kulve et al. 2018) or where similar trends were observed but not statistically significant (Toftum et al. 2018), the brightness sensation of participants in the current study was not affected by indoor temperature. However, results might have been confounded by the indoor illuminance levels that, even though within comfortable ranges, changed across experimental sessions.

Finally, indoor temperature did not affect brightness preference, overall visual environment comfort, lighting comfort, lighting pleasantness nor lighting relaxation. However, considering that such visual perception evaluations were not investigated in past studies on temperature–color interactions, it is not possible to compare results with other findings.

5.2. Cross-modal interaction effects of temperature and colored glazing on visual perception

Past studies on electric light investigating combined variations of thermal conditions and color (both of objects and of the light), led to findings in line with the HHH, with bluish objects and lights more positively evaluated in a warm thermal condition, and yellowish/orangish objects and lights more positively evaluated in a cold thermal condition. These results, associated to the concept of *alliesthesia* (Cabanac 1971), suggest a cross-modal interaction effect of temperature and color, with evaluations affected by both the levels of temperature and color evaluated. As a consequence, interaction effects were expected in this study in terms of color of the light evaluations.

However, when analyzing the entire dataset in the first part of the statistical analysis, interaction effects between temperature and colored glazing levels were not significant for the majority of visual perception evaluations. Only responses indicating the reason of visual discomfort were substantially affected by interactions, with participants indicating the color of the glazing as the reason of visual discomfort more often under the blue glazing at the low temperature (19 °C) and under the orange glazing at the high temperature (26 °C).

Despite not being statistically significant when the entire dataset was analyzed, an interaction effect between temperature and the orange and blue glazing was revealed by a visual inspection of the data. Further analyses at each temperature level retaining the responses under the orange and blue light only, confirmed the interaction. Results indicate that, only in the colder thermal condition (19 °C), the orange glazing resulted as being more comfortable and pleasant, and was chosen more often as first or second choice compared to the blue one. The difference between the two colors was not significant at 22 °C and 26 °C, as the scores of the evaluations for the blue glazing increased with temperature, while those for the orange glazing decreased (for color comfort and pleasantness) or did not vary across temperatures (for color ranking).

We hypothesize that a full interaction between colors and temperature levels would have been observed with the analysis of the full dataset with an additional warmer temperature level, as in our study the warmest temperature level investigated was not evaluated as "slightly warm" but only as "neutral" on a thermal sensation scale (Chinazzo et al. 2018a). We speculated that, at temperatures considered "slightly warm" or "warm" by participants, the blue glazing would result as more comfortable and pleasant than orange, and would have been chosen more often as first or second choice in the preference ranking. Our hypothesis is based on results of previous studies and would confirm the HHH-trend found in all of them (Kearney 1966; Kim and Hiromi 1998; Kim and Tokura 2001; Nakamura and Oki 2000; Wang et al. 2017): bluish colors were more positively evaluated at high temperatures and yellowish colors at low temperatures. The fact that participants complained the most about uncomfortable color of the glazing under blue color at the low temperature, and under orange color at high temperature further confirms this hypothesis.

In conclusion, despite not being confirmed at higher temperature levels, our results corroborate the presence of interaction effects between colors and temperatures, with results in line with the HHH. This finding suggests the bi-directionality of the HHH, with colors affecting the thermal perception and temperature influencing the visual perception of colors.

5.3. Unimodal effects of colored glazing on visual perception

As for the analyzed studies in the literature, colored glazing did affect the visual perception of the transmitted daylight, both in terms of color of light and visual environment evaluation. However, our results differ quite substantially from those of past investigations.

With reference to the color of the light evaluations, the neutral glazing came out as the most comfortable, acceptable, pleasant, and preferred compared to the blue and the orange ones. The neutral glazing appeared also to be the most comfortable and pleasant when the visual environment was evaluated, in terms of light quantity (brightness comfort), overall visual environment comfort and light quality (lighting comfort). These results therefore differ from those of past studies indicating the glazing with a warm-shift as the preferred one (Arsenault et al. 2012; Cuttle 1979; Liang et al. 2018; Vossen et al. 2016). This might be due to the saturated colors of the blue and the orange glazing types used in the present study, and to the fact that the entire surface of the openings was coated with colors. This choice resulted in an extremely colored indoor environment compared with those of past investigations, using glazing with only slight color-shift or saturated colored glazing partially covering the glazed surface.

Differences in the color stimuli between the current and the past studies could also explain the different results related to the brightness sensation evaluations of the light quantity, as strongly saturated lights more strongly affect the reflectance and the luminance of the surfaces of the indoor space, which in turns might influence the evaluation of the light quantity. The results of this study indicated that daylight tinted by the orange glazing was perceived as the least bright in comparison to both blue and neutral glazing, whereas past studies reported an enhanced perception of brightness with bronze glazing (Arsenault et al. 2012; Boyce et al. 1992). It is also important to point out that illuminance conditions were not completely stable in this study due to the use of daylight, and that the measured illuminance levels under the orange glazing were on average lower compared to those under the blue and the neutral ones (Table 4). Considering that the investigation of the effects of colored glazing on brightness perception was not the primary focus of this experiment, we suggest that further studies should address this problem with a different experimental design.

Referring to visual perception evaluations not examined in past studies on windows with a slight color-shift, we reported that the orange glazing resulted as the most relaxing in comparison to the blue and the neutral ones. Moreover, it was perceived as the warmest, followed by the neutral and the blue ones, in line with the established psychological association between colors and temperature in western culture.

5.4. Study limitations

Some limitations need to be considered for the generalizability of the reported findings. First, as previously anticipated, illuminance conditions were not totally stable across experiments and during each session, although they were always within comfortable ranges. Some results, therefore, might have been affected by the change of illuminance levels, especially those pertaining to light quantity evaluation (i.e., visual sensation, comfort, and preference). As experiments with daylight will always present this challenge, it would be beneficial if further investigations addressing the same research questions were conducted again, with illuminance conditions kept as constant as possible (e.g., experiments conducted with clear sky conditions only). Second, the study did not reach uncomfortably warm thermal conditions and hence interactions between color and temperature were not significant when the full dataset was analyzed. Additional investigations involving uncomfortably warm thermal conditions are recommended to confirm the suggested hypothesis about full interaction, with orange glazing preferred under cold temperatures and blue glazing preferred under warm conditions. Third, cultural and age differences may also have played a role in the obtained results, and investigations with a different population could further help the understanding of temperature–color interactions when daylight is investigated. Finally, experiments were controlled under laboratory conditions and

future studies should verify that similar results occur in real conditions. Other experimental investigations in a controlled environment are anyhow suggested to test the effects of additional variables that were fixed in the current experiment but that might influence visual perception evaluations such as the quality and quantity of the view to the outside (e.g., would the reported effects be the same with a mountain view?).

6. Conclusions

This study described the first investigation on the effect of combined variation of temperature (19 °C, 22 °C, and 26 °C) and daylight transmitted through saturated colored glazing (blue, orange, and neutral) on visual perception of daylight. The goal was to determine whether cross-modal effects of indoor temperature on visual perception can be observed when evaluating "colored" daylight (i.e., daylight transmitted through glazing with a saturated color), and to estimate the unimodal effects of saturated glazing on visual perception of daylight.

Expanding the results of past investigations under electric light conditions, this study confirms the cross-modal effect of indoor temperature on occupant visual perception when daylight and colored glazing are involved. In particular, cross-modal *main* effects are observed for the visual environment evaluation, and cross-modal *interaction* effects (i.e., interactions between the levels of temperature and glazing color) for the color of the light evaluation. With reference to the former effect, the visual environment was affected by indoor temperature in terms of brightness comfort and of the so-called lighting cold-warm association, independently of the color of the glazing. First, higher brightness comfort scores were reported in the colder thermal condition (19 °C) in comparison to the warmer one (22 °C). Then, the light was associated with warmer adjectives at the higher thermal conditions (22 °C and 26 °C) in comparison to the coldest one (19 °C). In terms of cross-modal interaction effects of temperature and glazing color type, results show that only at the lowest temperature level (19 °C) did the blue glazing lead to a less comfortable and pleasant light than the orange glazing. Also, only in this cold condition, the blue glazing was chosen less than the orange one. Such cross-modal interaction effects, although not confirmed at uncomfortably warm thermal conditions, do confirm results of past investigations, and corroborate the HHH and its bi-directionality (i.e., colors affect thermal perception and temperature influences visual perception of colors).

Unimodal effects of saturated colored glazing were significant for all the visual perception evaluations. However, as opposed to results of past studies on glazing with less saturated colors, the neutral glazing was preferred and it was considered more comfortable and pleasant compared to the orange and the blue ones. Differences might be due to the strong saturation of the colors used in this study and to the fact that the entire glazed surface was colored. Finally, the orange glazing came out as the most relaxing one and the color-warmth association was confirmed, with orange glazing indicated as the warmest and the blue one as the coldest.

The specific results of this study expand the existing knowledge on temperature–color interactions, extending results referring to the use of electric lighting to the evaluation of daylight transmitted through colored glazing. These conditions are becoming more realistic than the use of colored electric light due to the use of smart windows or of glazed surfaces with colored filters applied for esthetic or other reasons. The presented results in terms of cross-modal effects of temperature on visual perception indicate that whenever spectrally selective glazing is used, indoor thermal conditions should be considered to provide a comfortable indoor visual environment. More specifically, in a warm environment orange glazing should be avoided, as well as blue glazing in a cold thermal condition.

Results in terms of unimodal effects of colored glazing on visual perception can be useful for product selection in terms of color characteristics and distribution on the façade, as they indicate that the visual perception of saturated colors distributed on the entire glazed surface was different from that suggested in other studies investigating less saturated colors or different façade coverings. Care should be taken for the design and installation of smart window technologies that undergo

changes in their color if we want to guarantee visual comfort and satisfaction of building users, but also if we want to mediate specific feelings such as relaxation and apparent warmth.

More generally, the results of this study highlight and confirm the importance of considering the indoor environment as a dynamic and complex system with multiple interactions between factors, and not as composed by different and independent stimuli. As a consequence, a multidisciplinary and holistic approach should be adopted when the indoor environment is considered, in both research and practice.

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